Maitri Maribasayyanavar

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PROFESSIONAL SUMMARY

Automation & Robotics Engineering student with practical experience in embedded systems, sensor and actuator interfacing, robotics hardware integration, and troubleshooting. Skilled in Python and C, with project experience involving Arduino, Raspberry Pi, vision-based automation, and robotic end-effector design using SolidWorks. Interested in developing embedded systems, electronics for robotics, and hardware-software integration.

TECHNICAL SKILLS

Programming: Python, C, MySQL (Database Management Systems)
Embedded Systems: Raspberry Pi, Arduino, Microcontrollers, Sensor Interfacing, Actuator Interfacing
Electronics & Hardware: Wiring, Soldering, Multimeter, Oscilloscope, Hardware Assembly, Troubleshooting
Robotics: ROS 2 (Robot Operating System), URDF, Gazebo, RViz2, TF Transforms, Cobots
Computer Vision: OpenCV, Image Processing, Real-Time Vision Systems, Colour-Based Classification.
Industrial Automation: Industrial Automation, Omron Automation, Factory I/O, Sensor and Actuator Integration
Design & Simulation: SolidWorks, CAD Modelling, MATLAB, Simulink
Professional Skills: Problem Solving, Troubleshooting, Technical Documentation, Team Collaboration

EDUCATION

KLE Technological University, Hubli, Karnataka — **2027 (Expected)**
B.E. in Automation & Robotics — **CGPA: 8.15/10**

Vidyaniketan PU Science College, Hubli, Karnataka — **2023**
Pre-University Course (Science), Karnataka State Board — **90.50%**

S R Bommai Rotary Public School, Hubli, Karnataka — **2021**
CBSE — **87%**

PROJECTS

Design and Development of Magnetic End-Effector for TM5-700 Cobot
Tools: SolidWorks, CAD Modelling, Engineering Calculations
- Designed a magnetic end-effector in SolidWorks for pallet emptying and ASRS redocking with a collaborative robot.
- Calculated payload and magnetic force to determine the gripping capacity required for reliable pallet handling.
- Authored a technical paper on the magnetic end-effector design for pallet handling and ASRS redocking with the TM5-700 cobot [submitted to ICCIS 2026].

Vision-Guided Dual-Arm Robotic System for Cooperative Manipulation and Assembly *Ongoing*
Tools: MyBuddy 280, ROS 2, URDF, RViz2, Python, OpenCV
- Developing a vision-guided dual-arm system on MyBuddy 280 for cooperative object manipulation and assembly.
- Built the ROS 2 simulation workflow using URDF robot models, joint states, TF transforms, and RViz2 visualization.

Automated Vision-Based Flower Sorting System
Tools: Raspberry Pi, OpenCV, Python, Sensors
- Built a real-time flower classification system on Raspberry Pi with OpenCV, sorting by colour and quality.
- Integrated camera-based image processing with sensors and an automated sorting mechanism.

CERTIFICATIONS & WORKSHOPS

- **ROS Workshop** – Hands-on training in Robot Operating System (ROS) and robotic system integration.
- **Omron Workshop** – Hands-on training in industrial automation and robotics.
- Participant, 24-Hour National Level AI Hackathon (Yugma TechFest 2.0) and CodeHer Hackathon.
- **MATLAB Onramp** – Simulink & Control System Design.

LEADERSHIP & EXPERIENCE

- Run an Abacus franchise: train students, manage schedules, and oversee daily operations.
- Coordinate staff, process payroll, and created a website for our pre-primary school and maintain the progress through it.

ACHIEVEMENTS & ACTIVITIES

- Represented KLE Technological University at South Zone Inter-University Cricket Tournament, Chennai (2025–2026).
- Second Place, Inter-Department Table Tennis and Volleyball Competition.